

Lab-01 Cheatsheet — Set · Tuple · Dict + the 2 tasks

CSC4226 Artificial Intelligence and Expert System

At a glance

	List []	Tuple ()	Set {}	Dict {k:v}
Ordered	yes	yes	no	yes
Mutable	yes	no	yes	yes
Duplicates	yes	yes	no	unique keys
Access	<code>l[0]</code>	<code>t[0]</code>	<code>x in s</code>	<code>d["k"]</code>

Set — unique, unordered

```
s = {1,2,2,3}      # {1,2,3}      set() # empty ({ } is a dict!)
s.add(4); s.discard(9) # discard=safe, remove=errors
A|B  A&B  A-B  A^B      # union, intersect, diff, sym-diff
```

Tuple — immutable

```
t=(10,20,30); t[0]; t[-1]; t[1:] # index/slice
x,y = (4,5)      # unpack      one=(5,) # 1-elem needs comma
t.count(2); t.index(3)      # only 2 methods
```

Dict — key->value

```
d={"name":"Joy"}; d["name"]
d.get("city","N/A")      # no KeyError
d["age"]=22; d.update({...}); d.pop("age"); del d["name"]
for k,v in d.items(): ...
```

Task 1 — dedupe -> sort(desc) -> drop multiples of 3

```
n % 3 == 0      # divisible by 3      (!= 0 keeps non-multiples)
unique = set(numbers)
desc = sorted(unique, reverse=True)      # new list, any iterable
result = [n for n in desc if n % 3 != 0]
```

Task 2 — matrix multiply (A m*n . B n*p -> m*p)

```
# rule: cols(A) == rows(B); cell = dot product of row i, col j
result = [[0]*cols_B for _ in range(rows_A)] # NOT [[0]*c]*r (aliases!)
for i in range(rows_A):
    for j in range(cols_B):
        for k in range(cols_A):
            result[i][j] += A[i][k]*B[k][j]
# 1*7 + 2*9 + 3*11 = 58 ; 0(m*n*p)
```